

Technology Stack

Date: 15 July 2026

Team ID: XXXXXX

Project Name: Human Development Index (HDI) Prediction using Machine Learning

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, Bootstrap, JavaScript	Provides a responsive, interactive, and user-friendly interface for entering HDI indicators and displaying prediction results.
2	Backend / Server-Side	Python, Flask	Flask is lightweight and ideal for integrating machine learning models. Python provides powerful libraries for data analysis and prediction.
3	Database / Data Storage	MySQL	Stores user details, prediction history, country datasets, and system logs efficiently with structured data management.
4	Cloud / Hosting / Deployment	Render / Railway / Heroku	Enables easy deployment, scalability, and online accessibility with minimal configuration.
5	Version Control & CI/CD	Git, GitHub	Supports version control, collaborative development, code backup, and continuous project updates.
6	Third-Party APIs / Other Tools	Scikit-learn, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook	Scikit-learn is used for machine learning, Pandas and NumPy for data preprocessing, Matplotlib and Seaborn for visualization, and Jupyter Notebook for model development and testing.

Technology Stack Summary

The **Human Development Index (HDI) Prediction System** is developed using a modern and scalable technology stack. The frontend is built with **HTML, CSS, Bootstrap, and JavaScript** to provide an intuitive user interface. The backend uses **Python and Flask** to integrate the machine learning model and process prediction requests. **MySQL** is used for secure data storage, while **Git and GitHub** enable efficient version control and team collaboration. The machine learning model is implemented using **Scikit-learn**, with **Pandas, NumPy, Matplotlib**, and **Seaborn** supporting data preprocessing and visualization. The application can be deployed on cloud platforms such as **Render, Railway**, or **Heroku** for reliable online access.

Figure 5.1: Technology Stack of the Human Development Index (HDI) Prediction System



